

Co-Designing Mechanisms and Motion: Towards Safe and Deterministic Human–Robot Interaction

Durgesh Haribhau Salunkhe

Requested host institutions:

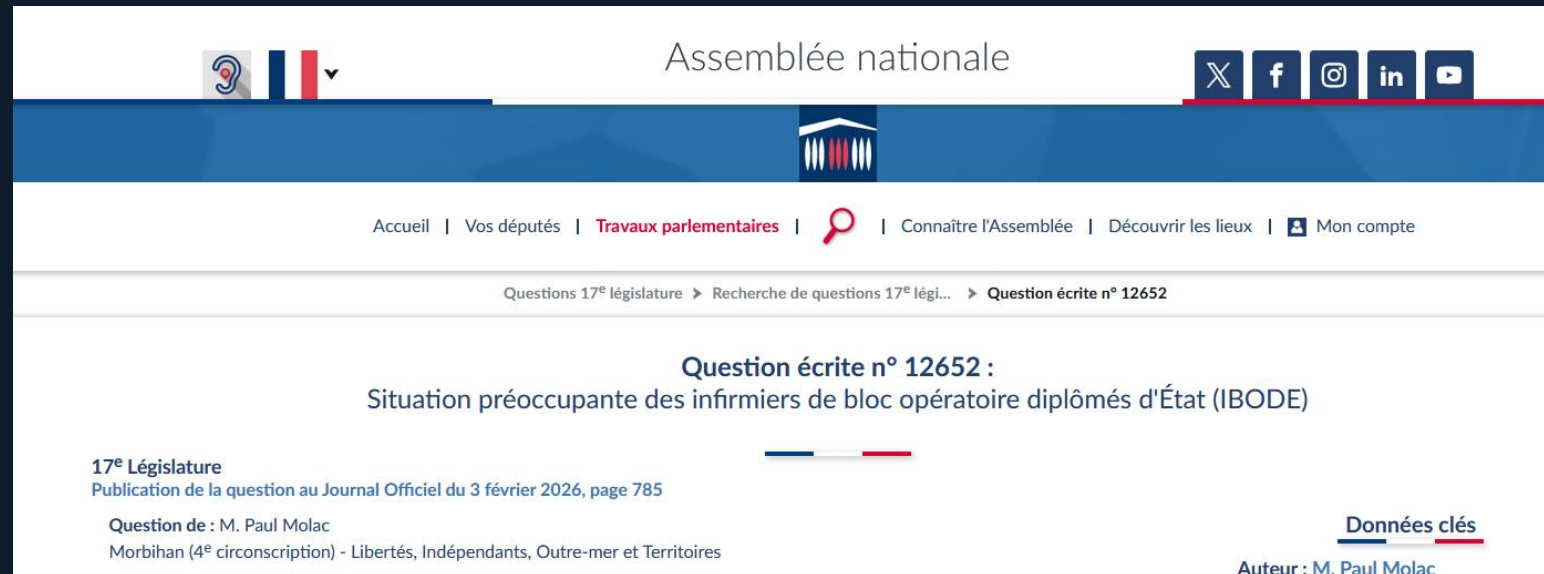
1. ICube, Strasbourg: Team RDH
2. LAAS, Toulouse: Team Gepetto



An urgent need!



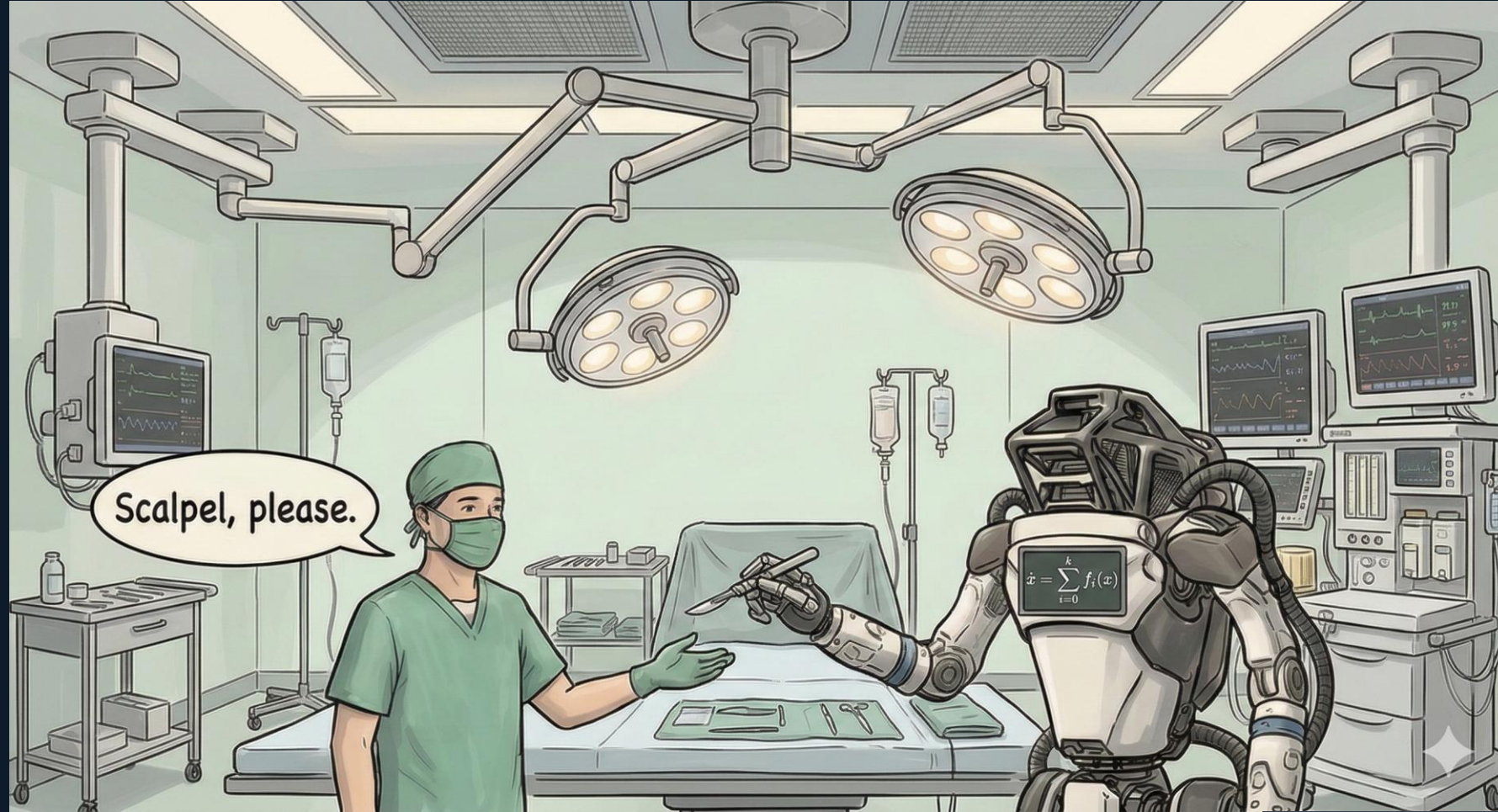
Le Parisien, 17 March, 2026



Question raised in the National Assembly, 03 Feb, 2026

Please pass me the scalpel

An example scenario



AI generated image

The three barriers to ubiquitous robots

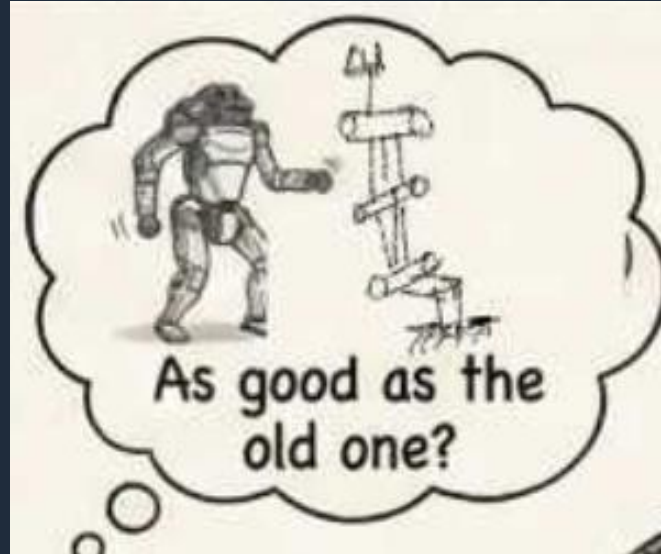
Why a surgeon may hesitate to trust a robot

Reliable Execution



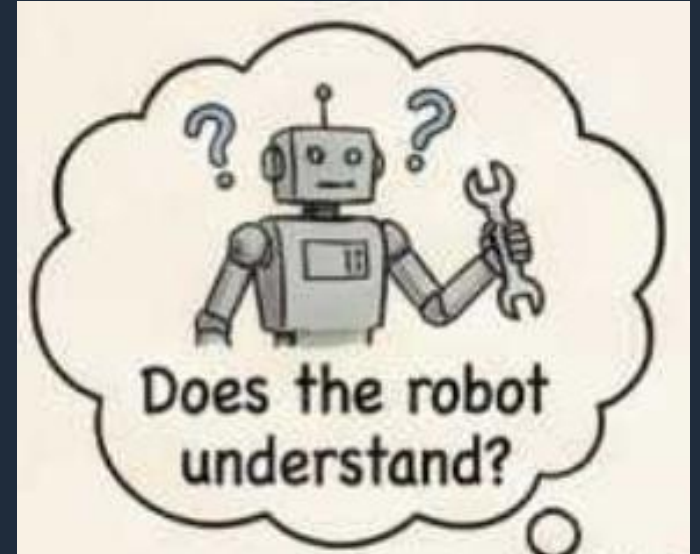
The real world is unstructured: robot execution must be safe and explainable.

Skill Scalability



As new robots are developed, skills need to safely transferred across all generations of robots.

Embodied alignment



Robot design and actions should be kinematically-grounded while working with LLM and GenAI

Research timeline

Timeline

Research Theme 1: Safe and Reliable Robot Behavior

1. Active force compliance in 6-RSS parallel platform

Journals: RCIM'19

Conferences: AIR'20

Software:

Research Theme 2: Task-Oriented Robot Design

1. Design optimization of parallel robots

Journals: MMT'20, JMD

Conferences: ICRA'22

Software: ParaOpt (used by LS2N, DFKI)



Director: Damien Chablat

Collaborations:

CHU Nantes, DFKI Germany

Research timeline

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Research Theme 1: Task-Oriented Robot Design

1. Design optimization of parallel robots
2. Kinematic analysis of generic 3R and 6R robots
3. Novel designs for Human-Robot Interaction

Journals: MMT'20

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Research Theme 2: Safe and Reliable Robot Behavior

1. Active force compliance in 6-RSS parallel platform
2. Path planning framework for 6R robots
3. Kinematic Intelligence for cross-robot skill transfer

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Collaborations with:
Sorbonne, Linz, Innsbruck

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Research Theme 2: Safe and Reliable Robot Behavior

1. Active force compliance in 6-RSS parallel platform
2. Path planning framework for 6R robots
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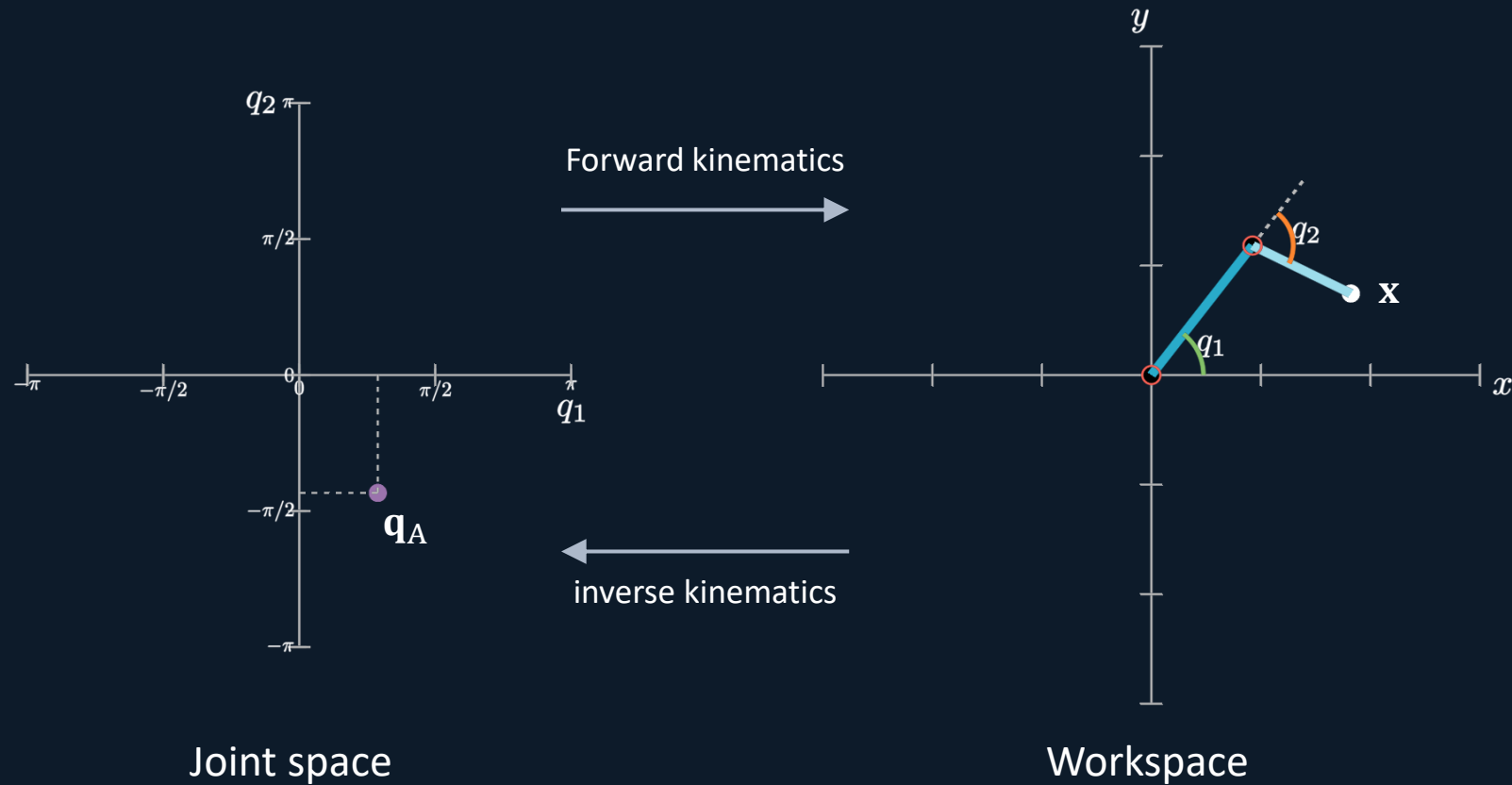
Supervisor: Aude Billard

Collaborations with:
IIT, India, IRI, Spain

Research activity

Doctoral work

nR robot: A robot with 'n' revolute joints. (I will use 2R, 3R, 6R and 7R)

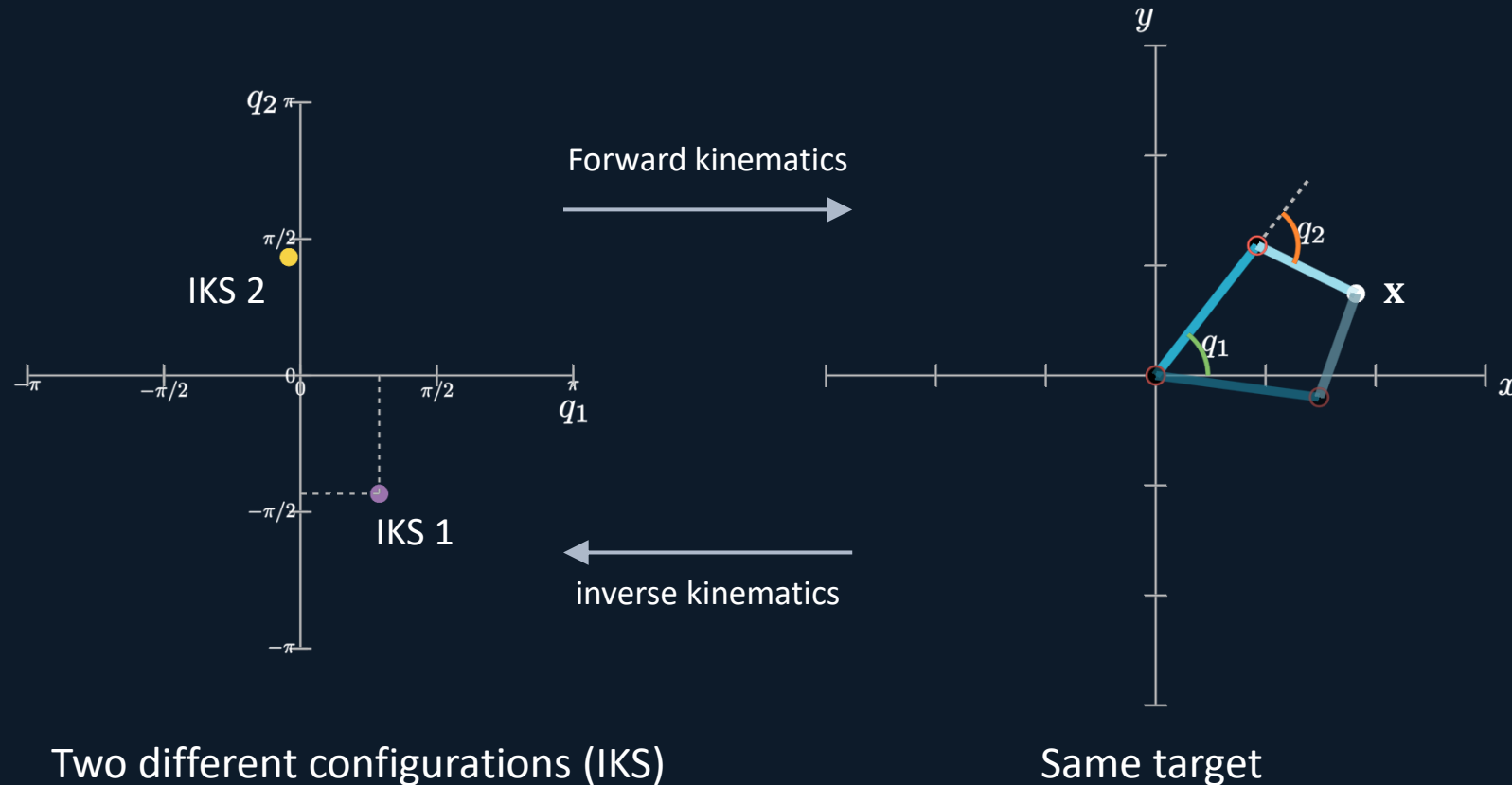


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Inverse kinematic solutions(IKS): different numbers of ways a robot can reach the same target



Research activity

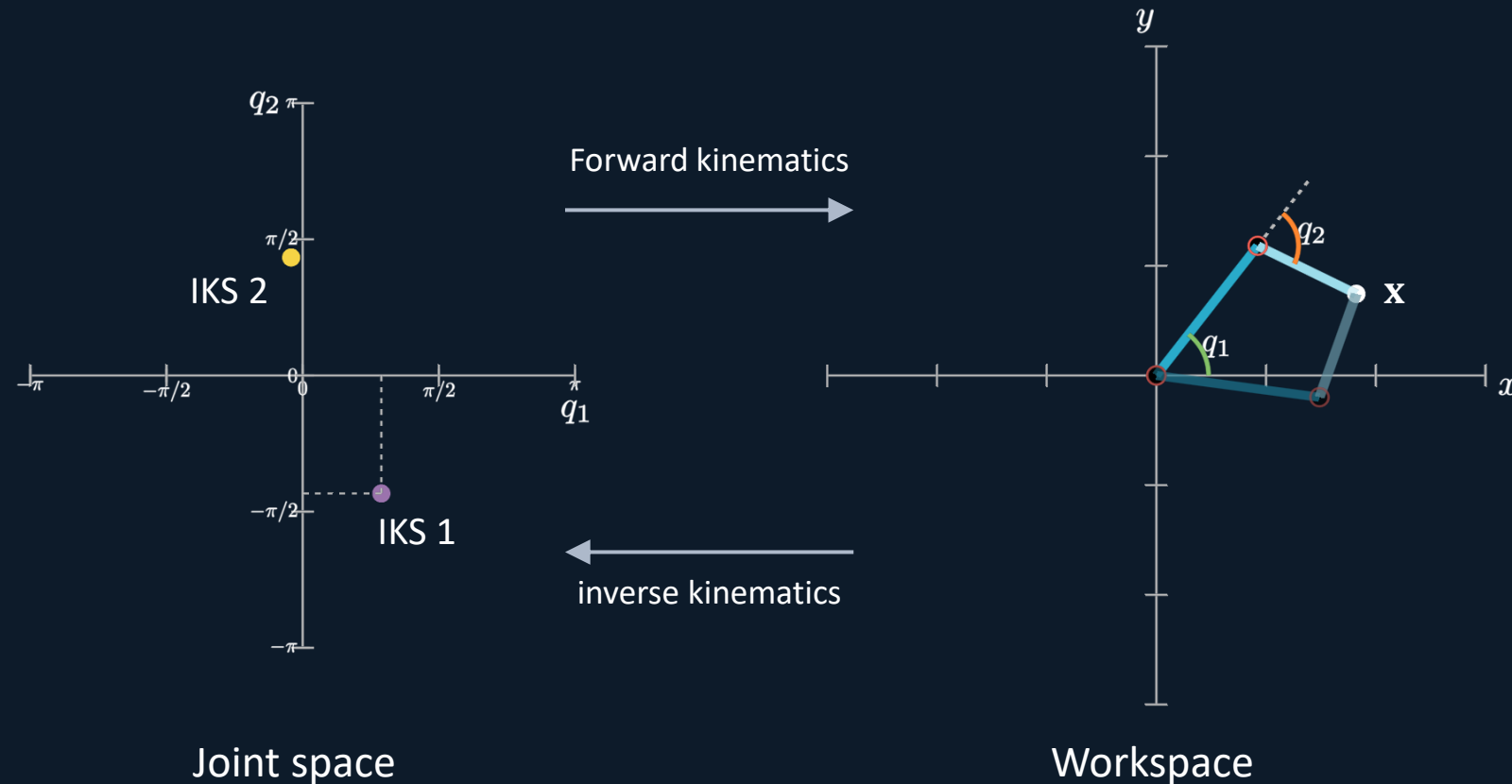
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***Jacobian:** maps the joint space velocities to workspace velocities*

$$\dot{\mathbf{x}} = \mathbf{J}(\mathbf{q}) \dot{\mathbf{q}}$$



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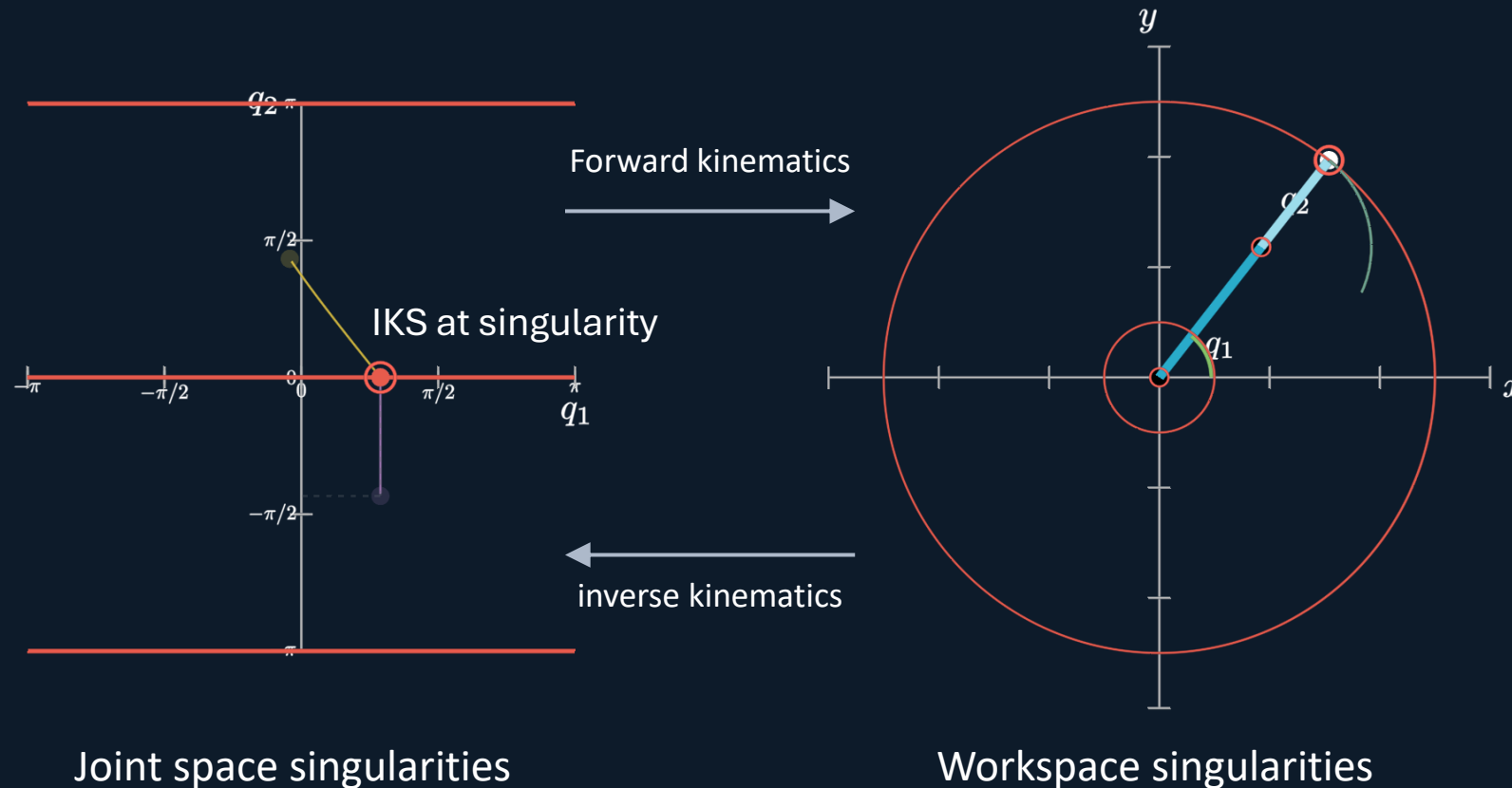
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Singularity: Configurations where Jacobian is not full rank

$$\det \mathbf{J}(\mathbf{q}) = 0$$



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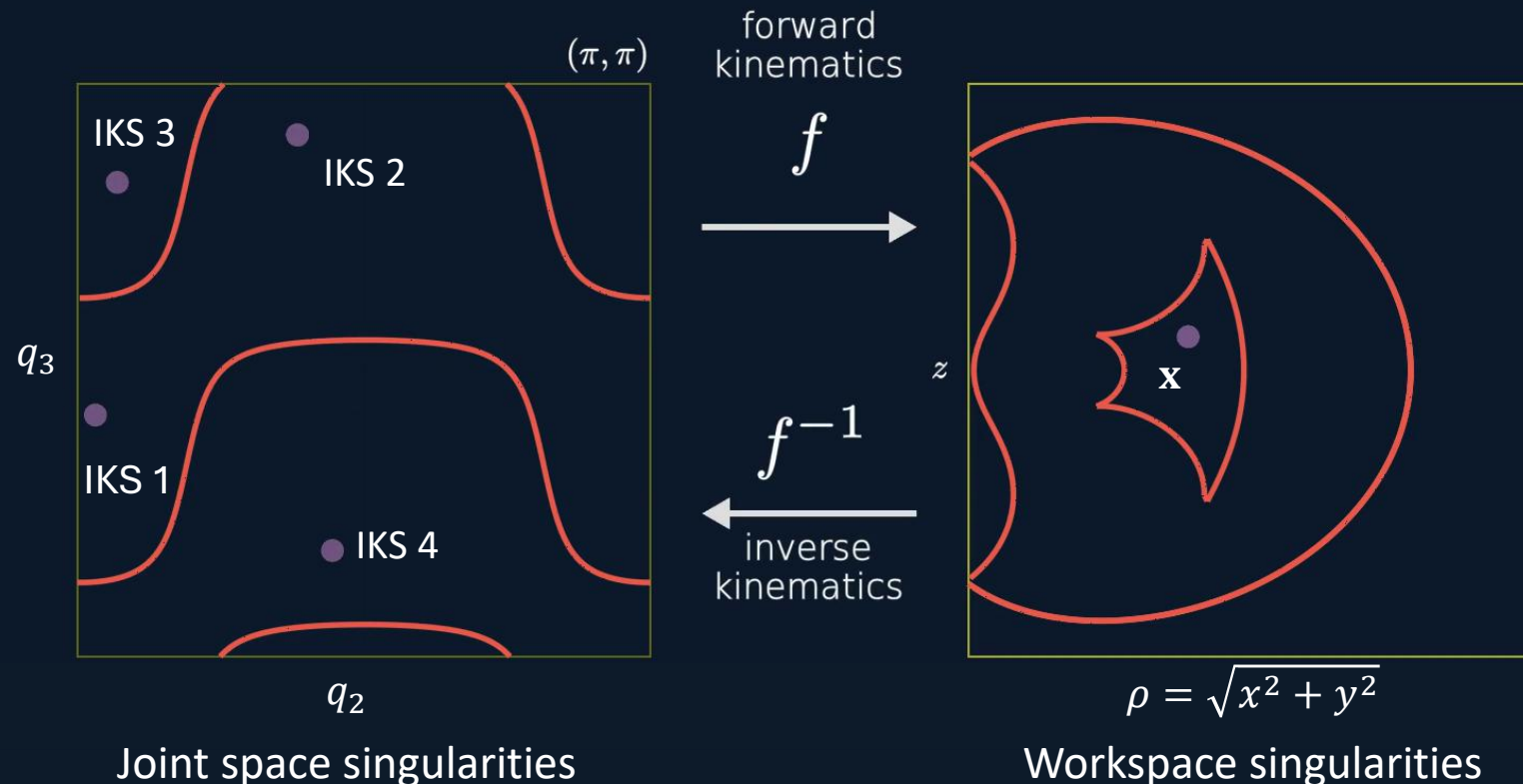
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Cuspidal Robots: Robots that change inverse kinematic solutions without crossing a singularity



Research activity

Doctoral work

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Inverse kinematic solutions(IKS): different numbers of ways a robot can reach the same target

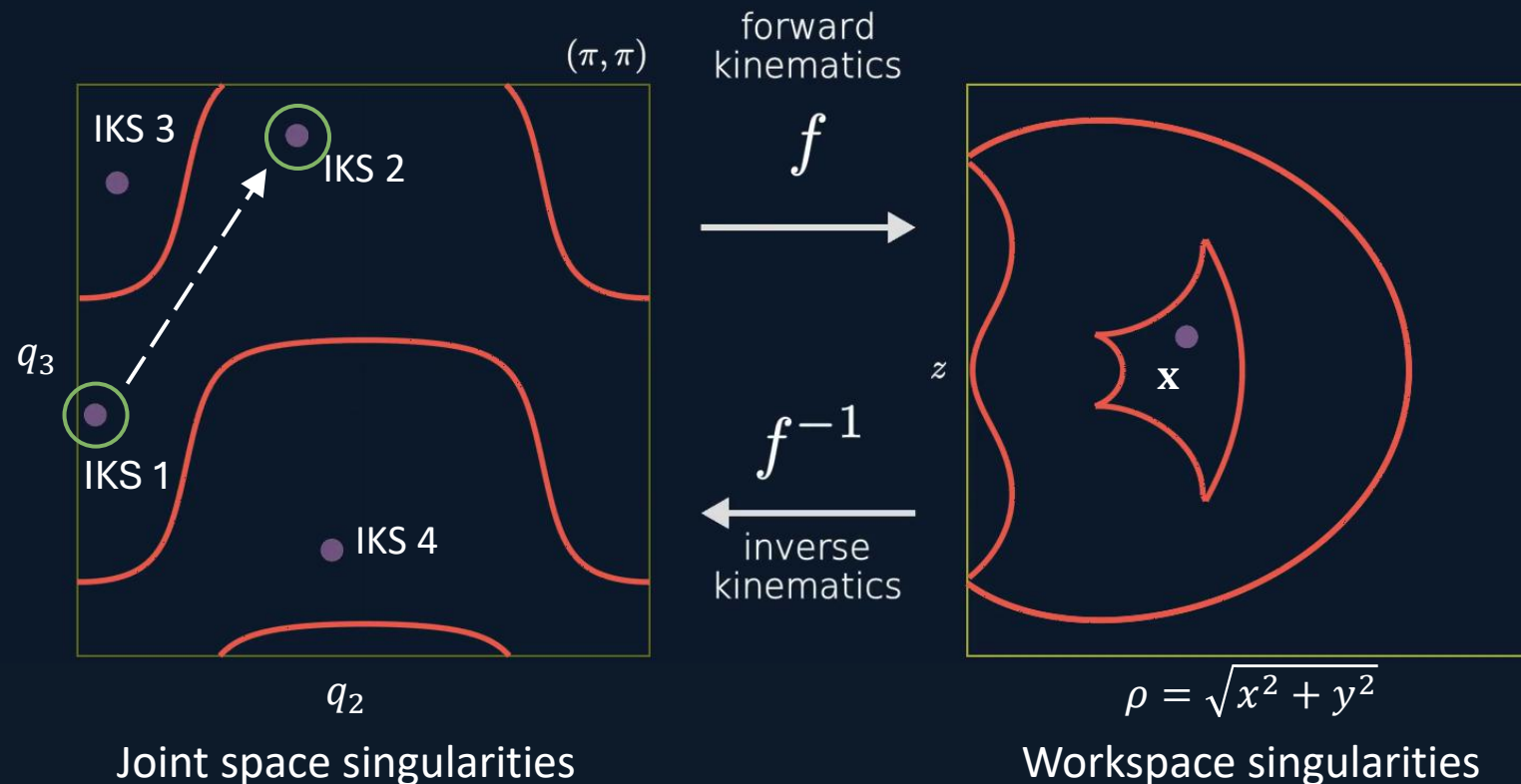
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Cuspidal Robots: Robots with multiple IKS in a singularity-free region in joint space

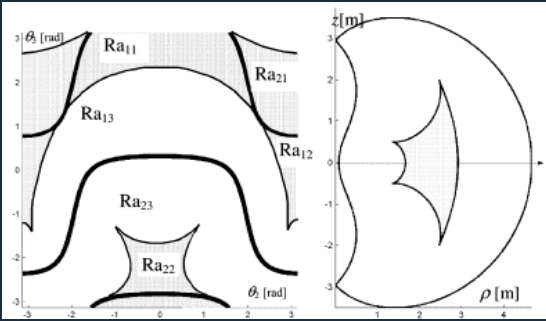


Research activity

Doctoral work

PREVIOUS WORK

- Trajectory planning issues in 3R cuspidal robots



(Wenger '04)

- Cuspidality analysis of 6R robots



Is a noncuspidal robot

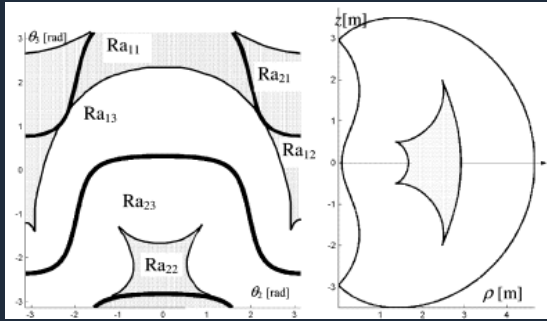
(Capco et al. '20)

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1. How to determine if a given 6R robot is cuspidal or not?

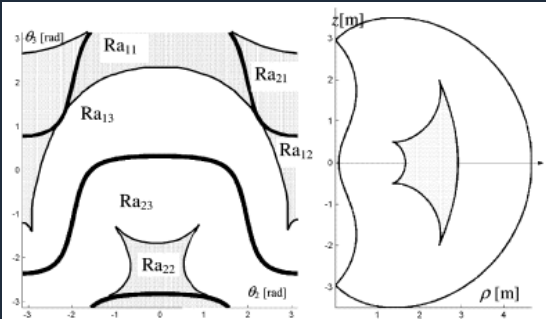
2. What are the implications on path planning for a 6R cuspidal robot?

Research activity

Doctoral work

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Is a noncuspidal robot

(Capco et al. '20)



Certified IK + Certified connectivity
=
Deciding cuspidality

Result 1:

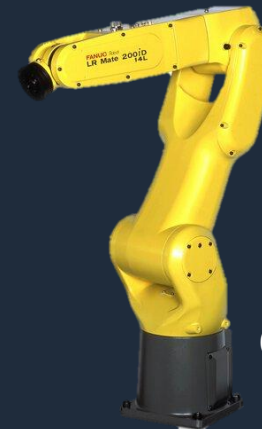
Even a small deviation from conventional design almost always leads to cuspidal robots.

(ISSAC'22, IJRR'24)

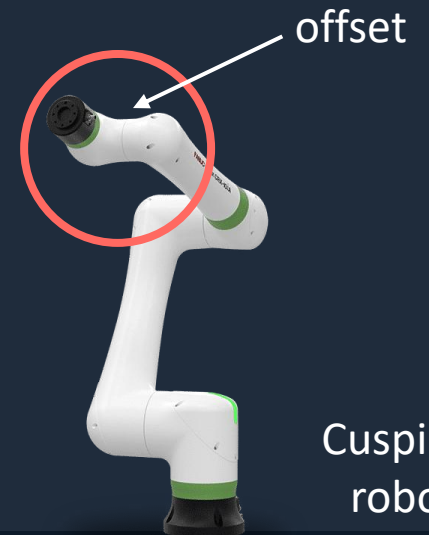


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2. What are the implications on path planning for a 6R cuspidal robot?



Conventional robot



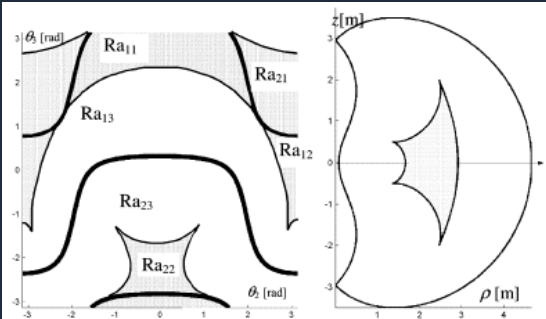
Cuspidal robot

Research activity

Doctoral work

PREVIOUS WORK

- Trajectory planning issues in 3R cuspidal robots



(Wenger '04)

- Cuspidality analysis of 6R robots



Is a noncuspidal robot

(Capco et al. '20)



Perform workspace analysis and explore different path properties for cuspidal robots.

Result 1:

Even a small deviation from conventional design almost always leads to cuspidal robots.

(ISSAC'22, IJRR'24)

Result 2:

A framework for safe and reliable path planning of commercial 6R robots.

(ICRA'23, IJRR'24)



1. How to determine if a given 6R robot is cuspidal or not?

2. What are the implications on path planning for a 6R cuspidal robot?

Research activity

Postdoctoral work

PREVIOUS WORK

Morphology-aware transfer

1. Infer the robot morphology (Trabucco et al. '22)
2. Hint the learning model about robot structure with biases (Sferrazza et al., '24)
3. Train a single policy that handles robot variations (Mu et al, '26, Yang et al., '26)

Research activity

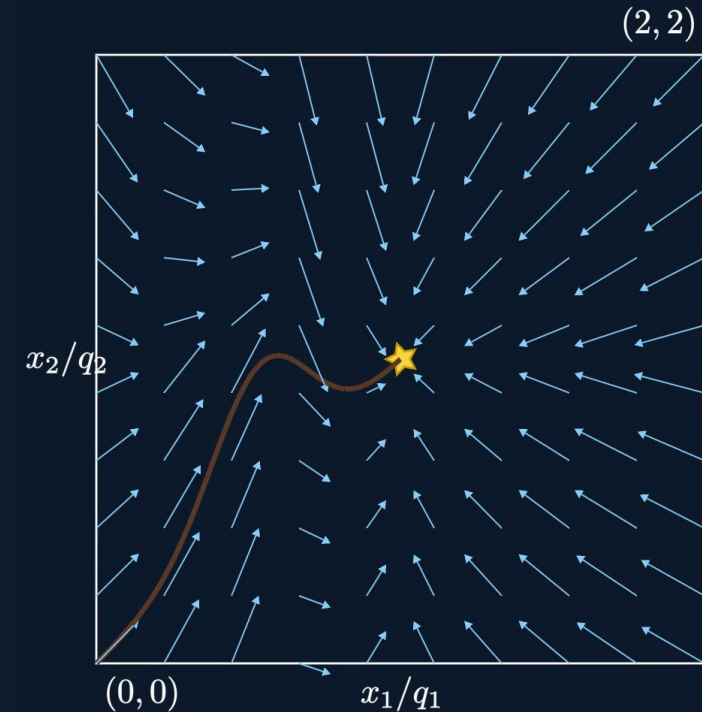
Postdoctoral work

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Control policy: maps the robot's current state to its next action or velocity command $\dot{s} = f(s)$



A control policy can be visualized as a vector field in the parameter space

Research activity

Postdoctoral work

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1. Explainable transfer of a user's demonstrated behaviour across different robots.

Research activity

Postdoctoral work

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1. Explainable transfer of a user demonstrated behaviour across different robots.
2. Modern approach for kinematic analysis



Embedding robot structure into learning for safe and deterministic motion.

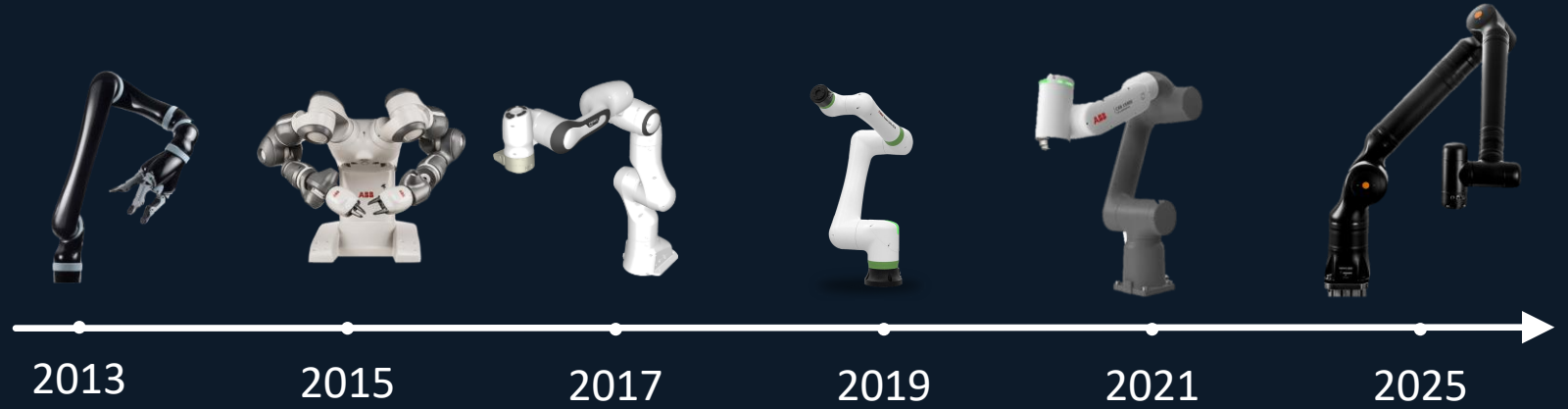
Stable robot-agnostic policy	Stable DS
+	+
Deterministic robot-specific execution	Singularity analysis
=	
Explainable cross-robot skill transfer	

1. Introduced a novel paradigm for an explainable cross-robot skill-transfer framework ([Science Robotics '26](#))
2. Collaboration with IRI Spain on kinematic analysis using Conformal Geometric Algebra ([IMA '25 x2](#))

Phase 1

Short-term scientific focus

Reliable Execution



1. Many modern commercial robots (6R and 7R) are cuspidal.
2. Almost all of them are used for collaborative purposes.

Analysis of cuspidal robots is both urgent and essential.

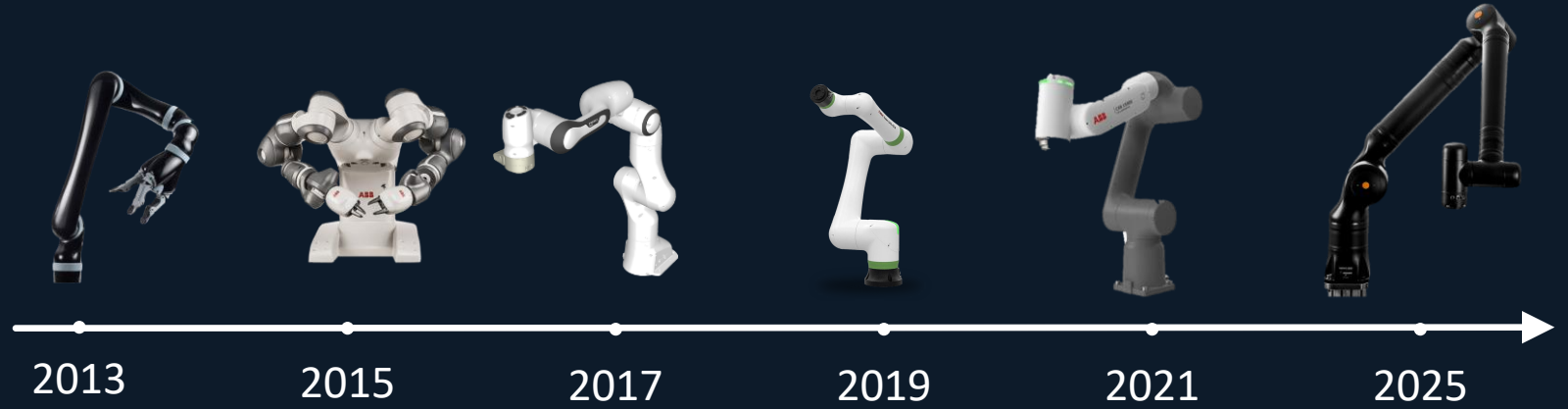
Phase 1

Short-term scientific focus

Reliable Execution



1. Lack of analytical results for kinematic analysis of cuspidal robots.
2. Identical tasks can admit multiple behaviors, undermining deterministic and repeatable execution.



1. Many modern commercial robots (6R and 7R) are cuspidal.
2. Almost all of them are sold under the category 'cobot'.

This makes the development of a safe, deterministic control-policy framework for cuspidal robots both urgent and essential.

Phase 1

Short-term scientific focus

Reliable Execution



1. Lack of analytical results for kinematic analysis of cuspidal robots.
2. Identical tasks can admit multiple behaviors, undermining deterministic and repeatable execution.

Co-design approach:

1. Perform global kinematic analysis for cuspidal robots.



1. Use accelerated processing and efficient algorithms for workspace analysis

Efficient
IK solvers

+

Reduced-dimension
analysis

+

Certified cell
decomposition

=

Global kinematics
of cuspidal robots

Conformal
Geometric
Algebra

Redundant angle
parameterization

Cylindrical
Algebraic
Decomposition

Phase 1

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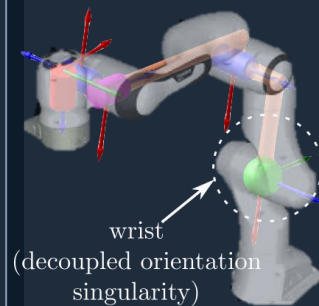
Global kinematics
of cuspidal robots

Conformal
Geometric
Algebra

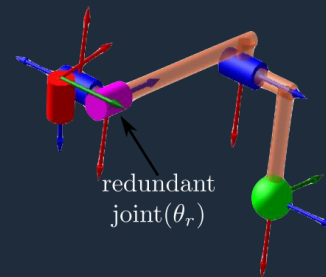
Redundant angle
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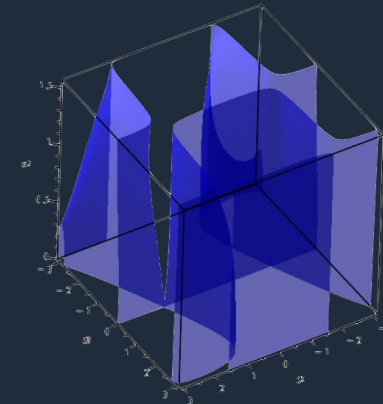
example
↓



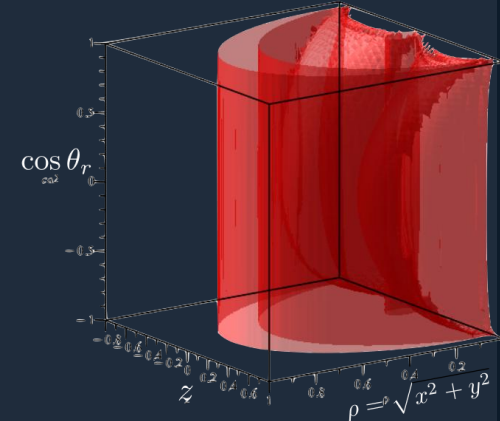
7R robot



4R + 3R robot



Stacked joint space
singularities



Stacked workspace
singularities

Phase 1

Short-term scientific focus

Reliable Execution

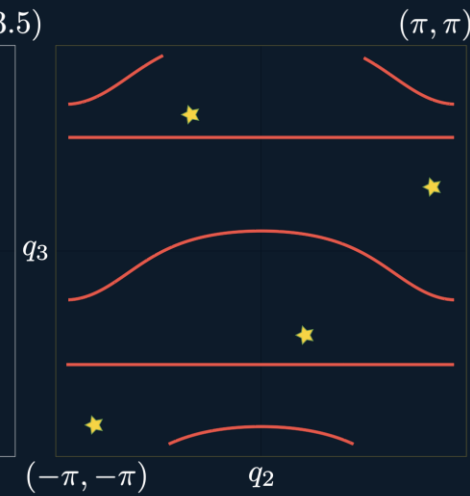


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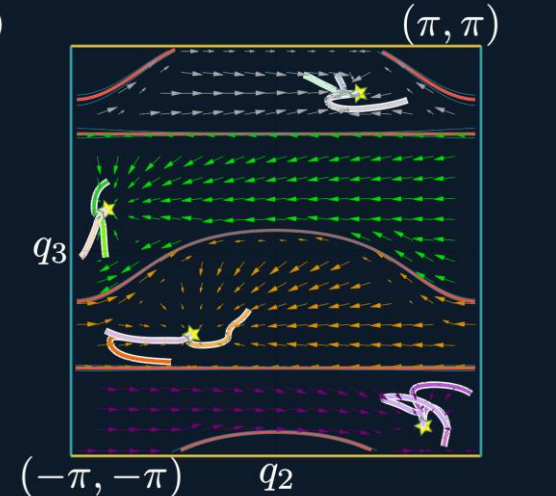
Noncuspidal robot



Workspace
'star' is an attractor



Joint space
'star' is IKS of attractor



Joint space policy
Individual policy in each region

Phase 1

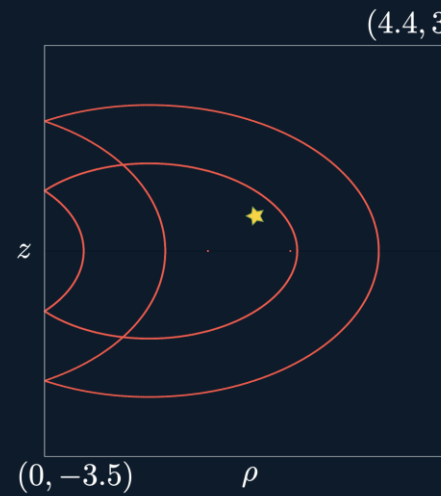
Short-term scientific focus

Reliable Execution

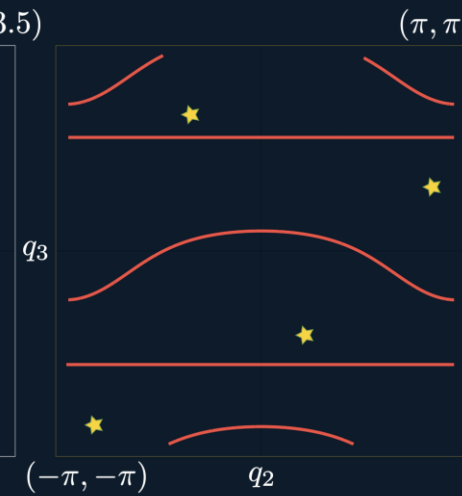


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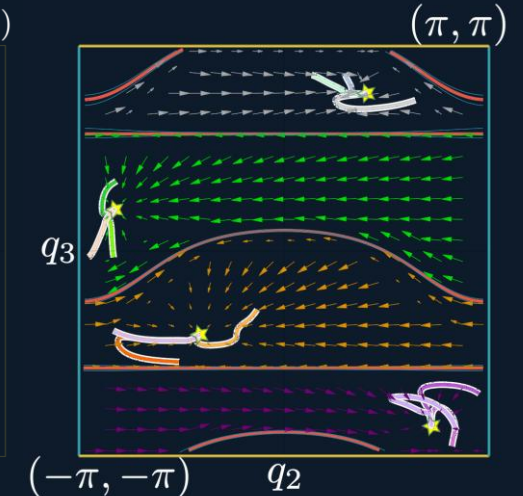
Noncuspidal robot



Workspace
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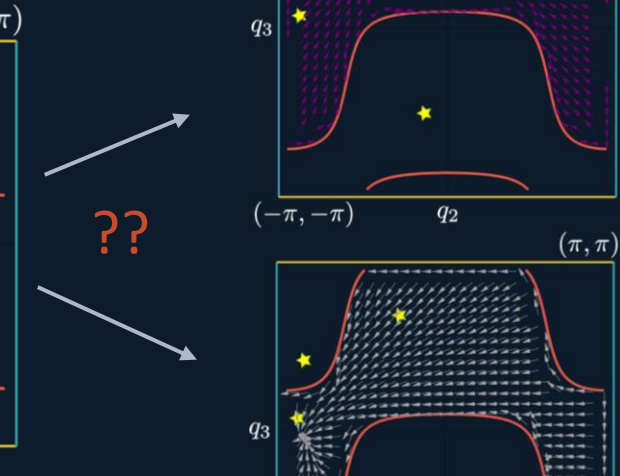
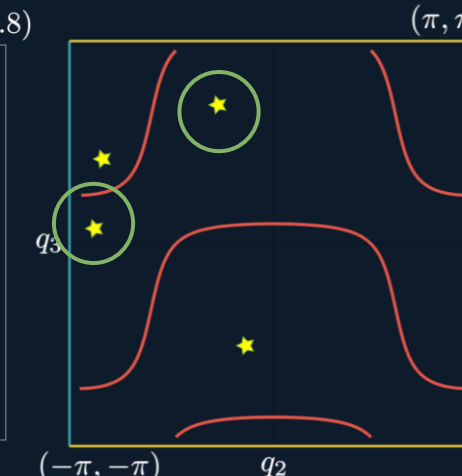
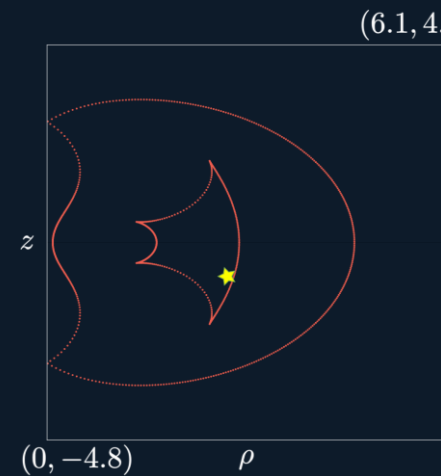


Joint space
'star' is IKS of attractor



Joint space policy
Individual policy in each region

Cuspidal robot



??

Phase 1

Short-term scientific focus

Reliable Execution



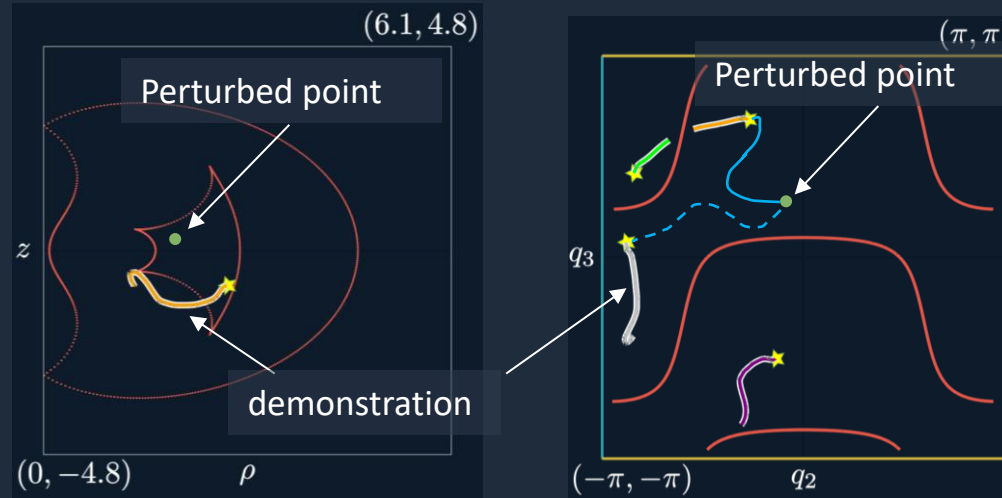
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Co-design approach:

1. Perform global kinematic analysis for cuspidal robots.
2. Develop metrics to identify the suitable control policy

2. Study a metric to enquire proximity to user intent



Which path do I choose?

The solid blue or dashed blue?

Phase 1

Short-term scientific focus

Reliable Execution



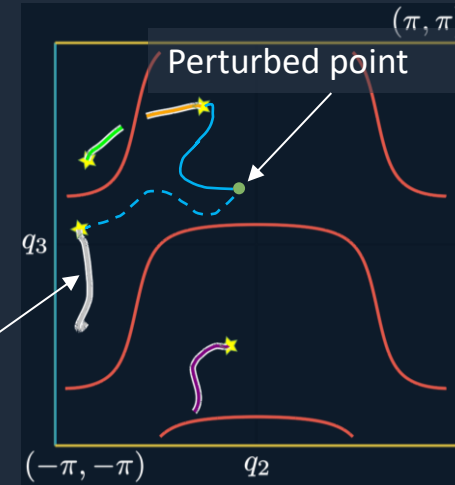
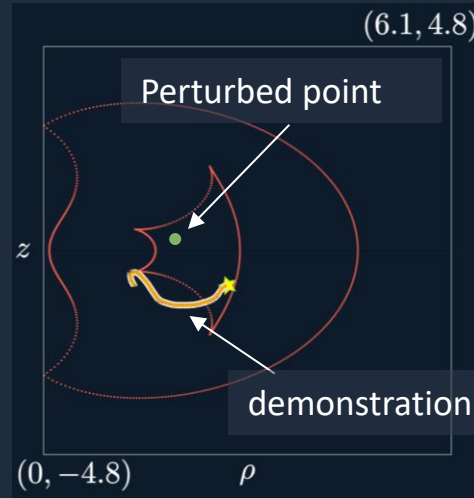
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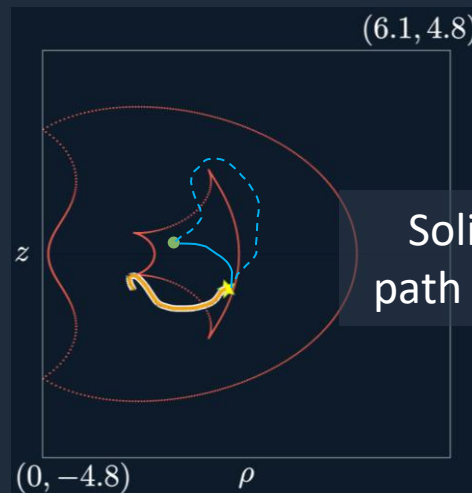
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Which path do I choose?

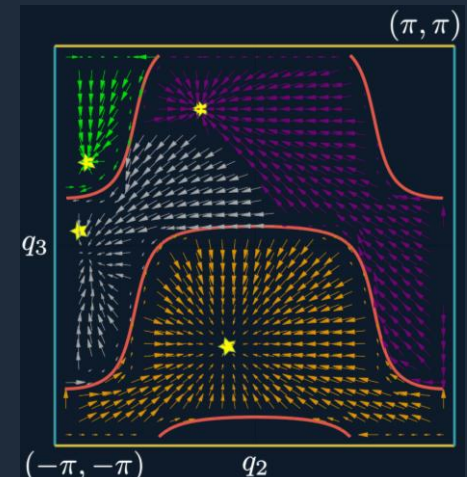
The solid blue or dashed blue?

Use fast oneshot forward mapping and measure the deviation from the intended behaviour



Solid blue path is better

Expected final control policy



Phase 2

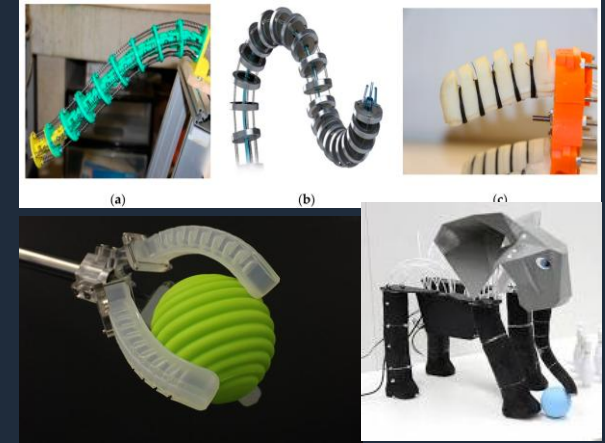
Mid-term scientific focus

Skill Scalability

Rigid links



Soft material



Increasing flexibility

Increasing precision

Phase 2

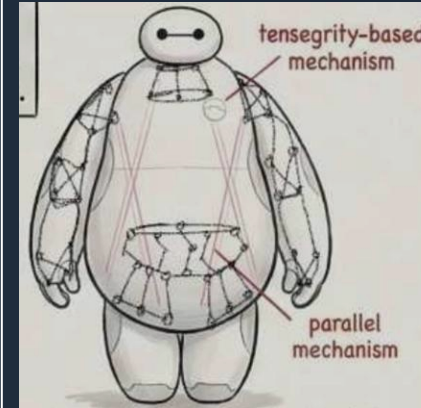
Mid-term scientific focus

Skill Scalability

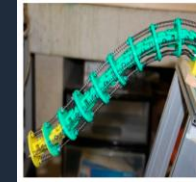
Rigid links



Robots for HRI



Soft material



Increasing flexibility

Increasing precision

Properties desired in human-robot interaction (Haddadin and Croft, '16):

1. Mechanical impedance
2. Lightweight designs
3. Transparency (natural motion)

Phase 2

Mid-term scientific focus

Skill Scalability



1. Robots are either precise and rigid or soft and compliant.
2. Explainable skill transfer framework that works across present robots and caters for future designs is missing.

Co-design approach:

1. Rigid–soft trade-off through novel joints and mechanisms



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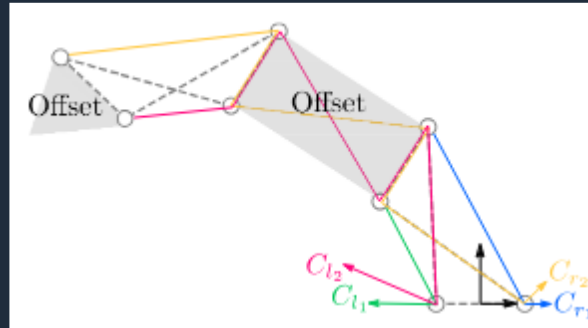
Mechanisms that
mimic human joints

+

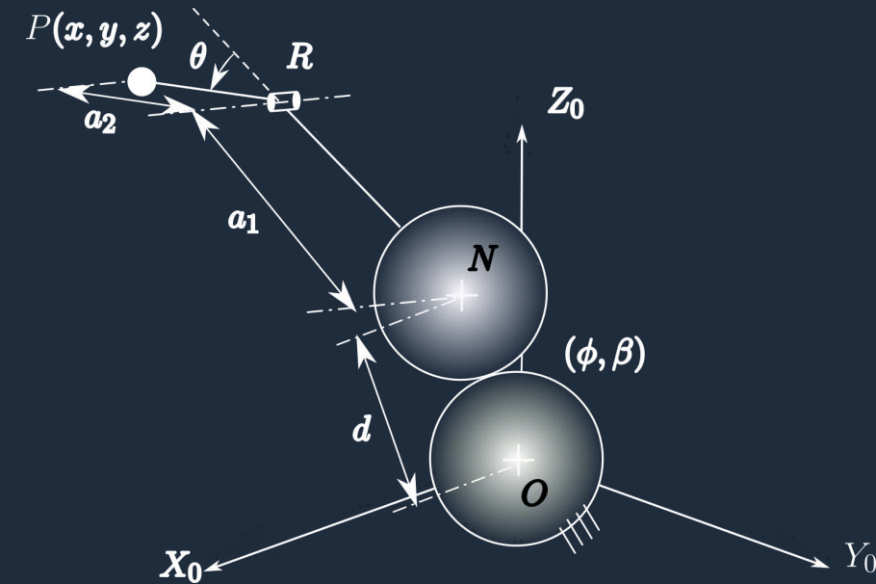
Cable-driven
actuation

=

Light robots with
variable stiffness



Cable driven anti-parallelagram joints
(Muralidharan and Wenger, '25)



Serial arms with sphere-on-sphere
rolling kinematics

Explore hybrid designs and novel actuations to maximize the desirable properties
for HRI

Phase 2

Mid-term scientific focus

Skill Scalability



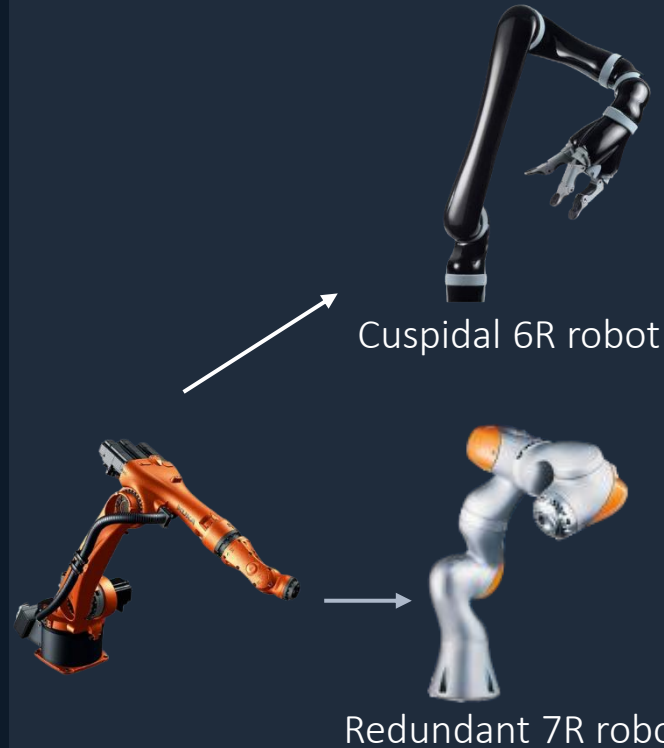
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Co-design approach:

1. Rigid-soft trade-off through novel joints and mechanisms
2. Explainable skill transfer through Kinematic Intelligence

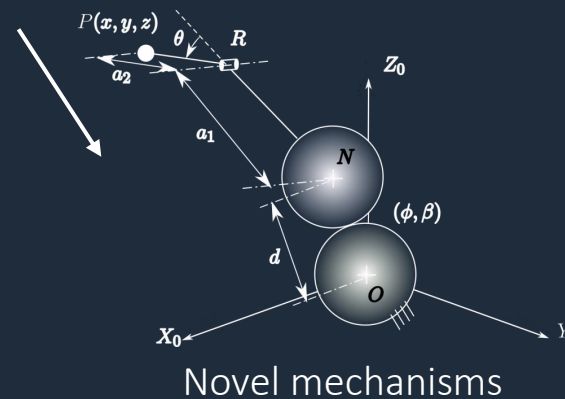


2. Explainable skill transfer through Kinematic Intelligence

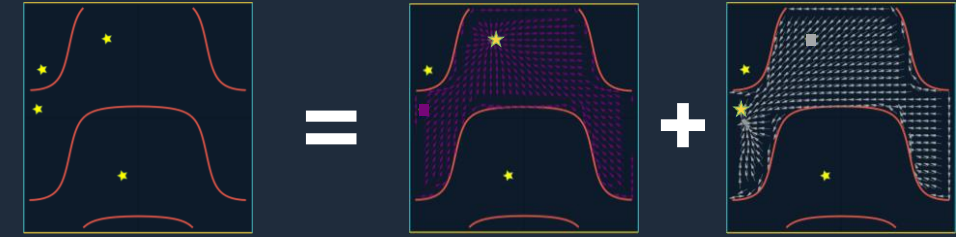


Cuspidal 6R robot

Redundant 7R robot

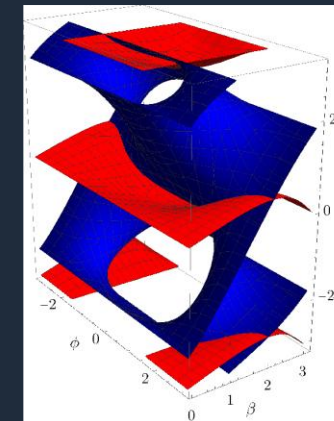
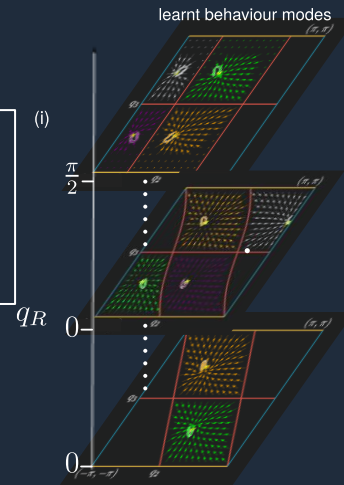


Novel mechanisms



Treat cuspidal robot as a collection of noncuspidal robot for each IKS

Parameterize the control policy with respect to the redundant angle



Study topological characteristics of singularities

Choice of host laboratories



Team RDH

Le laboratoire des sciences de l'ingénieur, de l'informatique et de l'imagerie (UMR7357) (iCube)

Strasbourg



Team Gepetto

Laboratoire de recherche spécialisé dans l'analyse et l'architecture des systèmes (UPR 8001) (LAAS)

Toulouse

Parallel robots

Medical robotics

Human-robot
interaction

Complements team's existing expertise

Pierre Renaud

Florent Nageotte

Hassan Omran

Kinematic
intelligence

Path Plannning

Design Optimisation

I bring my research expertise

Geometric Algebra

Learning from
demonstration

Transfer learning

I contribute in expanding the current
expertise of the team

Olivier Piccin

Nicolas Padoy

Infrastructure

1. Collaborative robots, redundant robots (x 2) (Fig. A)
2. Haptic teleoperating system (Fig: B)
3. Strong collaboration with medical teams (IHU + Camma) (Fig. B, C)

Perfect environment to extend my research in medical applications



A



B



C

Path planning

Optimisation

Transfer learning

Complements team's existing expertise

Philippe Souères

Nicolas Mansard

Olivier Stasse

Kinematic
Intelligence

Cuspidal robots

Parallel robots

I bring my research expertise

Co-design

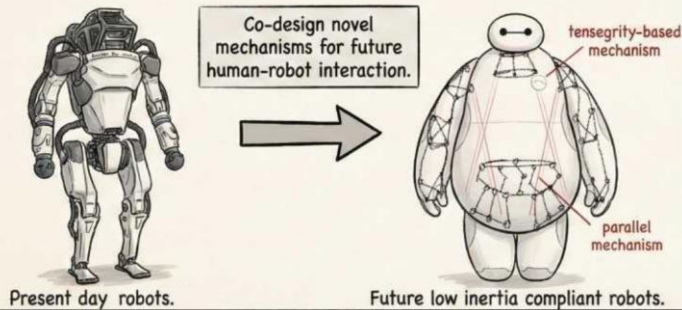
Learning from
demonstration

I contribute in expanding the current
expertise of the team

Embodied Safety

Phase 1: Year 0-2

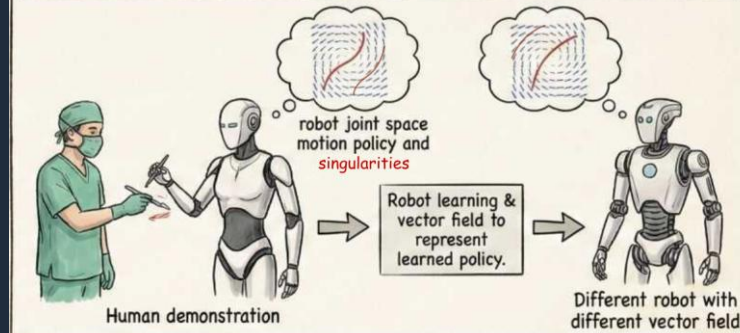
Key 1: Hybrid design, low inertia, tensegrity-based designs for safe HRI.



Skill Scalability

Phase 2: Year 2-5

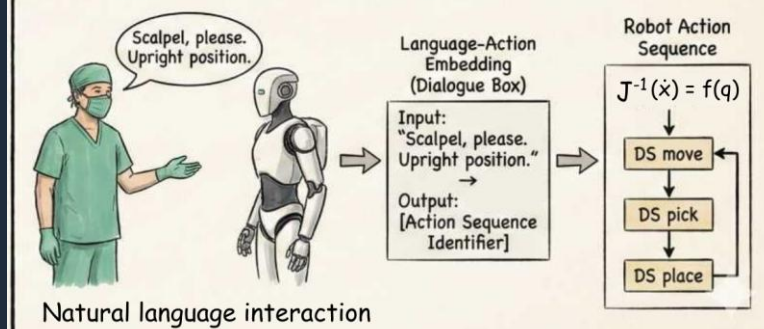
Key 2: Kinematic intelligence for Cross-robot skill transfer



Grounded Interaction

Phase 3: Year 5+

Key 3: Language-action embedding



Cuspidal robots
singularities and control
policy ambiguity

Skill transfer among current
robots and future-ready
new mechanisms

LLM based interactions need to
be analytically grounded for
motions and designs

Use modern tools to study
workspace

Develop metrics to
disambiguate control policy

Develop new mechanisms
fit for HRI

Kinematic Intelligence for
current and future robots

Explore analytical coupling of
Dynamical systems with language

Embed kinematic validation for robot
designs that use Generative AI

Phase 3

Long-term scientific focus

Grounded Interaction



1. Deterministic embedding of language with motion is essential
2. Designs developed using generative models need kinematic validation



Co-design approach:

1. Develop motion language framework by mapping language tokens with DS
2. Develop generative models that are aware of kinematic analysis.

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